SIMATS ENGINEERING

Department of Computer Science and Engineering

CSA15 Cloud Computing and Big Data Analytics

CO2 AT2 - VM Configuration and Security Evidence Workbook

Guided practical workbook for Azure VM, local VM, Docker container, security checks, recovery evidence and GitHub submission

Assessment Metadata

Course Code	CSA1512
Course Title	Cloud Computing and Big Data Analytics
Assessment	CO2 AT2 - VM Configuration and Security Evidence
Submission Type	Individual digital workbook based on the same capstone title used in CO1 and CO2 AT1
Faculty	Dr C. Rajesh Babu
Employee ID	SSETSCS415
Submission Mode	Completed workbook, evidence screenshots and GitHub repository link through Google Classroom

How This Workbook Works

- Use the same capstone title carried from CO1 and CO2 AT1.
- Read each procedure block, perform the action, verify the checkpoint and then fill the response table.
- Use Azure first when account activation is available. Use the local VM path when Azure is blocked. Use Docker for container comparison. Use simulation only when practical deployment is blocked.
- Screenshots must show the student's actual configuration or verified simulation work. Hide credentials, keys, tokens and sensitive account details.
- Upload the completed workbook and evidence to GitHub, then submit the GitHub repository link through Google Classroom.

Virtualization Tracks for CO2 AT2

Azure is the preferred cloud path. Local VM and container tracks are included for hands-on resilience.

Track A - Azure Linux VM
Preferred cloud evidence: subscription, VM, SSH, NGINX/test service, firewall, shutdown

Track B - Local VM
VirtualBox or VMware with Tiny Core, Alpine, Debian netinst or Ubuntu Server

Track C - Docker / Podman
Run a lightweight service, map a port, inspect logs, compare with VM

Track D - Simulation
Used only if deployment is blocked; must include technical configuration and security evidence

Required common output: completed workbook, screenshots, command outputs, security choices, README.md, GitHub repository link and GCR submission.

Technical Orientation

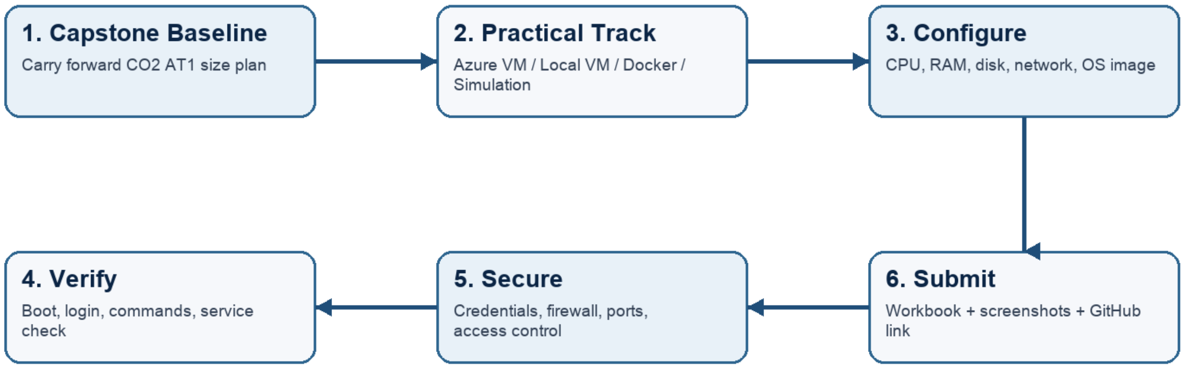
The practical work is not limited to clicking through screens. Each checkpoint requires observation: what appeared on screen, why it appeared, what it proves and how it connects to the capstone infrastructure plan.

Part 1: Student and Capstone Baseline

Student Name	D.S.THARUNIYA PRIYA
Register Number	192511317
Class / Slot	Slot B
Capstone Title	PulsePay : Cloud based fake payment Detection
CO2 AT1 Recommended VM Size	vCPU: 2-4 RAM: 12 GB Disk: 40 GB
Selected Practical Track	**Azure VM** / Local VM / **Docker** / Simulation
GitHub Repository Link	https://github.com/tharu-ui/Cloud-Computing-and-BDA/tree/main/PulsePay
Google Classroom Submission Date	29/07/2026

CO2 AT2 Practical Evidence Flow

Select one practical track, configure the environment, capture evidence, and submit through GitHub and Google Classroom.



Capstone Infrastructure Element	Carry-Forward from CO2 AT1
Workload type	AI/Python module + API backend hybrid
Main application modules	Screenshot upload, OCR extraction, ELA tamper analysis, ML fraud-risk scoring, merchant dashboard
Expected users / traffic assumption	1,000 registered merchants, ~200 concurrent peak, ~3.33 RPS
Database need	Firestore (managed NoSQL)
File storage need	Cloud Storage for screenshot images
AI/Python/analytics need	OCR (Vision API) + ELA + ML fraud classifier
Security concern	Screenshots may contain partial UPI/account details
Recovery concern	Rollback points needed before schema/model changes

Part 2: OS Image Selection Guide

Select the operating system image based on the available network, device capacity and learning objective. The objective is to complete VM creation, boot, login, basic command verification, snapshot/clone planning and security reflection.

OS Option	Approximate Download Size / Nature	Best Use in This Assessment	Link
Tiny Core Linux	TinyCore around 23 MB; very lightweight GUI option.	Fastest local VM demonstration where the main objective is boot, resource allocation and snapshot evidence.	https://distro.ibiblio.org/tinycorelinux/downloads.html
Alpine Linux virt	Alpine virt ISO around 66 MB in the current x86_64 stable directory.	Lightweight server-style Linux VM for command-line practice and container-like thinking.	https://dl-cdn.alpinelinux.org/alpine/latest-stable/releases/x86_64/

Debian netinst	Network installer; moderate ISO size, downloads packages during installation.	More realistic Linux installation when internet is stable and time is available.	https://www.debian.org/download
Ubuntu Server	Larger server ISO; familiar cloud VM image.	Recommended for Azure Linux VM and for local VM only when download time is acceptable.	https://ubuntu.com/download/server

Selection Guidance

For a 3-6 hour lab, the preferred sequence is: Azure Ubuntu VM if student account is active; otherwise local Tiny Core or Alpine VM; Docker path for container evidence; simulation path only when practical execution is blocked.

Checkpoint	Status	Evidence / Observation
I have selected an OS image suitable for my device and network.	[X] Yes [ ] Rework	Alpine Linux virt (66MB) for local VM; Ubuntu Server 24.04 LTS for Azure VM
I have recorded the official download page or cloud image name.	[X] Yes [ ] Rework	Alpine: dl-cdn.alpinelinux.org/alpine/latest-stable/releases/x86_64/ ; Azure: Ubuntu Server 24.04 LTS (Gen2) selected in Azure Marketplace
I have checked whether the ISO is small enough for the lab duration.	[X] Yes [ ] Rework	Alpine ISO (~66MB) downloaded and booted in under 2 minutes on lab laptop

Student Entry: Selected OS image, version, reason and link used

Alpine Linux virt 3.24.1 (x86_64) was selected for the local VM. It was chosen over Ubuntu Server because its ISO (~66MB) is far smaller and boots faster than Ubuntu's, making it more practical for a lab environment with limited time and device resources. Link: https://dl-cdn.alpinelinux.org/alpine/latest-stable/releases/x86_64/. For the Azure cloud VM, Ubuntu Server 24.04 LTS (Gen2) was used, since it is the recommended default image for Azure Linux VM deployment and provides a realistic, production-representative environment for the SSH and service verification steps.

Part 3: Track A - Azure Linux VM Guided Procedure

> **Preferred Cloud Path**
>
> Use Azure first when the student account is active. Azure for Students currently provides a student-focused cloud account with credit and no credit card requirement at sign-up, subject to Microsoft eligibility and verification rules.

Azure for Students: https://azure.microsoft.com/en-us/free/students

Azure for Students offer details: https://azure.microsoft.com/en-us/pricing/offers/ms-azr-0170p

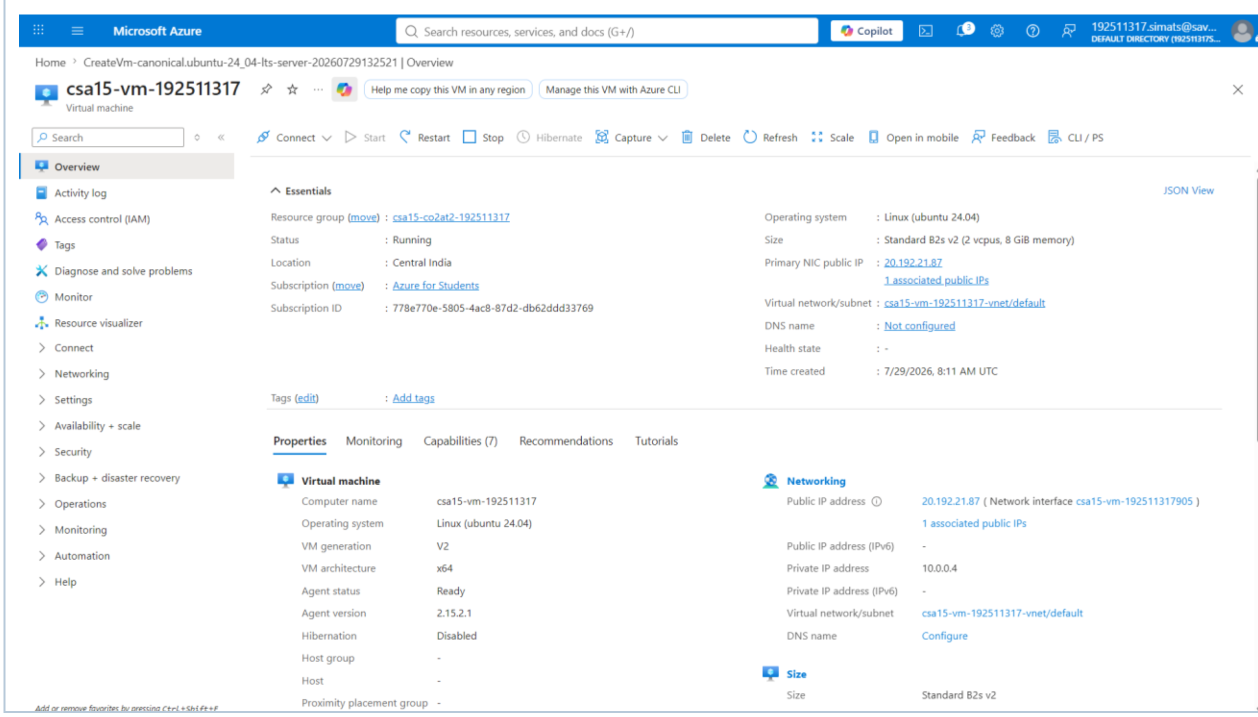
Azure Linux VM quickstart: https://learn.microsoft.com/en-us/azure/virtual-machines/linux/quick-create-portal

A. Account Activation and Portal Access

1. Open the Azure for Students page and choose the student sign-up option.
2. Sign in using the official student email account when available.
3. Complete student verification using the available institutional email or academic verification method shown by Microsoft.
4. Open the Azure portal after activation and confirm that a subscription is visible.
5. Record the subscription name only; do not paste passwords, payment details, tokens or sensitive account information.

Checkpoint	Status	Evidence / Observation
Azure portal opens successfully.	[X] Yes [ ] Rework	Portal loaded at portal.azure.com after student email verification through Azure for Students
A student or valid Azure subscription is visible.	[X] Yes [ ] Rework	"Azure for Students" subscription active and visible in account
The student understands that resources must be stopped or deleted after evidence capture.	[X] Yes [ ] Rework	Auto-shutdown configured for 11:00 PM IST daily; resource group will be deleted once grading is complete

Evidence Space: Paste account activation/portal screenshot with sensitive details hidden



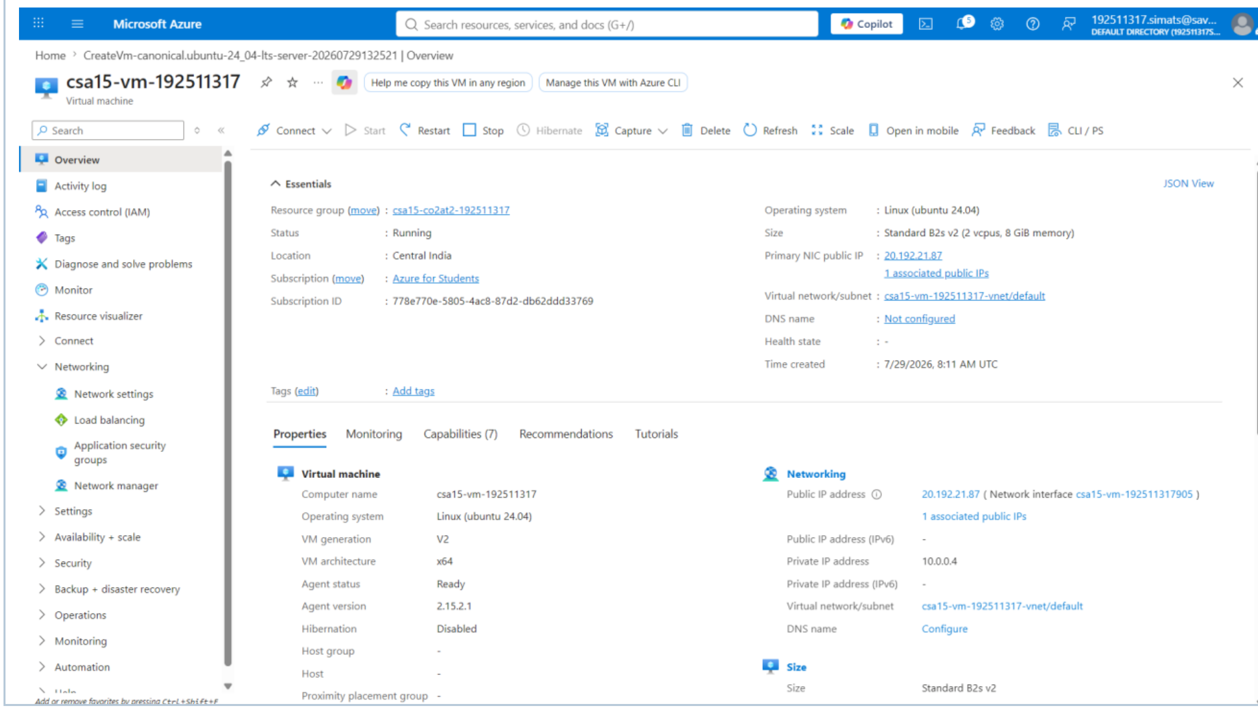
B. Create the Linux Virtual Machine

6. In Azure portal, search for Virtual machines and choose Create.
7. Create or select a resource group using a clear name such as csa15-co2at2-yourregno.
8. Enter a VM name such as csa15-vm-yourregno.
9. Select a nearby Azure region available in the subscription.
10. Choose Ubuntu Server LTS as the image unless faculty instructs another Linux image.
11. Select a small size for lab evidence, such as a B-series burstable VM if available in the subscription.
12. Select SSH public key or password according to faculty guidance. Do not share credentials in the workbook.
13. Allow SSH port 22 only when required. If a web service is tested, add HTTP port 80 only for the test window.
14. Review and create the VM. Wait until deployment succeeds.

Azure Setting	Student Value
Resource group	csa15-co2at2-192511317
VM name	csa15-vm-192511317
Region	Central India
Image	Ubuntu Server 24.04 LTS - Gen2
VM size	Standard B2s v2
vCPU	2
RAM	8 GiB
Disk type and size	Premium SSD LRS, OS disk (image default size)
Authentication method	SSH key
Open inbound ports	22 (SSH) — port 80 opened temporarily for NGINX test, then closed

Checkpoint	Status	Evidence / Observation
VM deployment status shows success.	[X] Yes [ ] Rework	Deployment validation passed; "Your deployment is complete" confirmation shown
VM overview page shows running status.	[X] Yes [ ] Rework	Status field on Overview page shows "Running"
Public IP or connection method is visible.	[X] Yes [ ] Rework	Public IP 20.192.21.87 visible on VM Overview page
The selected VM size is compared with CO2 AT1 sizing recommendation.	[X] Yes [ ] Rework	AT1 estimated 2-4 vCPU / 12GB RAM; deployed Standard B2s_v2 (2 vCPU / 8GB) — closely matches, slightly under the RAM estimate but sufficient for lab/evidence purposes

Evidence Space: Paste VM overview and configuration screenshot



C. Connect and Verify the VM

15. Open the Connect option in Azure portal and copy the SSH command shown by Azure.
16. Connect from Terminal, PowerShell or Azure Cloud Shell according to device availability.
17. After login, run: uname -a
18. Run: lsb_release -a or cat /etc/os-release
19. Run: free -h
20. Run: df -h
21. Run: ip addr or hostname -I
22. Record the output as screenshot evidence.

Verification Command	Expected Purpose	Student Observation
uname -a	Kernel and OS environment	Linux csa15-vm-192511317 6.17.0-1020-azure, x86_64 — confirms Azure-hosted Ubuntu kernel
cat /etc/os-release	Linux distribution details	Ubuntu 24.04.4 LTS (Noble Numbat)
free -h	RAM allocation	7.8Gi total, 511Mi used, 7.1Gi free — matches B2s_v2's 8GB allocation
df -h	Disk allocation	29GB root filesystem (/dev/root), 27GB available, 6% used
hostname -I	Private IP / network evidence	eth0 shows inet 10.0.0.4/24, matches Azure portal's private IP

Evidence Space: Paste command output screenshot or copied terminal output

```
csa15admin@csa15-vm-192511317:~$ uname -a
Linux csa15-vm-192511317 6.17.0-1020-azure #20~24.04.1-Ubuntu SMP Fri Jun 19 20:09:14 UTC 2026 x86_64 x86_64 x86_64 GNU/Linux
csa15admin@csa15-vm-192511317:~$ cat /etc/os-release
PRETTY_NAME="Ubuntu 24.04.4 LTS"
NAME="Ubuntu"
VERSION_ID="24.04"
VERSION="24.04.4 LTS (Noble Numbat)"
VERSION_CODENAME=noble
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=noble
LOGO=ubuntu-logo
csa15admin@csa15-vm-192511317:~$ free -h
               total        used        free      shared  buff/cache   available
Mem:           7.8Gi       511Mi       7.1Gi       4.0Mi       436Mi       7.3Gi
Swap:             0B          0B          0B
csa15admin@csa15-vm-192511317:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
/dev/root        29G  1.6G   27G   6% /
tmpfs           3.9G     0  3.9G   0% /dev/shm
tmpfs           1.6G  968K  1.6G   1% /run
tmpfs           5.0M     0  5.0M   0% /run/lock
efivarfs        128M   32K  128M   1% /sys/firmware/efi/efivars
/dev/sda16      881M   64M  756M   8% /boot
/dev/sda15      105M  6.2M   99M   6% /boot/efi
tmpfs           794M   12K  794M   1% /run/user/1000
csa15admin@csa15-vm-192511317:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 60:45:bd:ce:b3:a2 brd ff:ff:ff:ff:ff:ff
    inet 10.0.0.4/24 metric 100 brd 10.0.0.255 scope global eth0
```

D. Optional Web Service Check

23. Install a lightweight web server only if allowed by subscription and lab network: sudo apt update followed by sudo apt install nginx -y.
24. Start or verify the service: systemctl status nginx.
25. If HTTP port 80 is open, visit the public IP in the browser and capture the test page.
26. If port 80 is not opened, record the reason and keep SSH-only evidence.

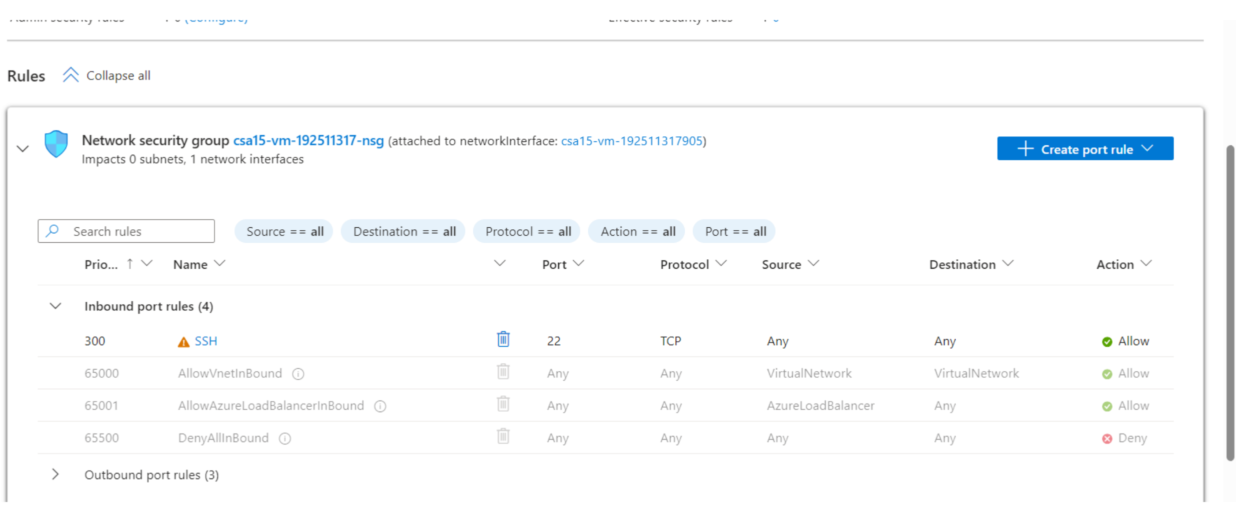
Checkpoint	Status	Evidence / Observation
The student can explain why a public port was opened or not opened.	[X] Yes [ ] Rework	Port 80 was opened temporarily only to browser-test the NGINX default page, then closed immediately after to minimize exposure
The service status or browser test was captured.	[X] Yes [ ] Rework	systemctl status nginx showed "active (running)"; browser test at http://20.192.21.87 displayed the NGINX welcome page
The public exposure was kept temporary for the assessment.	[X] Yes [ ] Rework	The port 80 inbound rule was deleted right after the browser test, restoring SSH-only (port 22) access

E. Azure Security and Cost-Control Closeout

27. Review Networking and record inbound port rules.
28. Confirm that no unnecessary public ports are open.
29. Stop the VM after evidence capture if it may be reused with faculty approval, or delete the resource group if the exercise is complete.
30. Capture stopped/deallocated or deletion evidence.
31. Record what would change for a production capstone deployment.

Security / Cost-Control Item	Student Evidence / Decision
Inbound ports allowed	SSH (22) only — port 80 (HTTP) was opened temporarily for the NGINX test and closed immediately
Authentication method used	SSH public key (RSA) — no password authentication
Public IP required?	Yes, temporarily — needed to SSH in and to browser-test NGINX; VM has Auto-shutdown enabled (11:00 PM IST daily) as a cost-control safety net
Stopped/deallocated or deleted?	Kept running (with Auto-shutdown scheduled) so faculty can verify the live deployment; will be deleted after review is complete
Reason for closeout choice	Faculty may want to independently verify the VM is genuinely deployed and functional, so it's being kept temporarily active rather than deleted immediately, with Auto-shutdown as a safeguard against runaway cost

Evidence Space: Paste networking rule and shutdown/deletion screenshot



Part 4: Track B - Local VM Guided Procedure

Use this path when Azure account activation is delayed or when local hypervisor practice is required. VirtualBox is recommended for broad accessibility. VMware Workstation/Fusion may be used where available.

VirtualBox downloads: https://www.virtualbox.org/wiki/Downloads

VMware Workstation and Fusion: https://www.vmware.com/products/desktop-hypervisor/workstation-and-fusion

A. Install the Hypervisor

32. Download VirtualBox or VMware only from the official website.
33. Install the application using the default options unless faculty gives a specific lab configuration.
34. Restart the system if the installer requests it.
35. Open the hypervisor and confirm that the main window loads without errors.
36. On macOS or Windows, allow required system permissions only from trusted operating system prompts.

Item	Student Entry
Hypervisor selected	**VirtualBox** / VMware Workstation / VMware Fusion
Version	7.2.14
Host operating system	**Windows** / macOS / Linux
Host CPU and RAM	13th Gen Intel Core i5-1334U, 15.7 GB RAM
Reason for selected hypervisor	VirtualBox was chosen because it's free, widely documented, and doesn't require a paid license like VMware Workstation — making it the most accessible option for a personal lab laptop

Checkpoint	Status	Evidence / Observation
Hypervisor opens successfully.	[X] Yes [ ] Rework	Oracle VM VirtualBox Manager opened with no errors
Student can locate New VM/Create VM option.	[X] Yes [ ] Rework	"New" button used to launch the VM creation wizard
Student has downloaded or selected a suitable OS image.	[X] Yes [ ] Rework	Alpine Linux virt 3.24.1 ISO downloaded and attached to the VM

B. Create the VM

37. Choose New VM or Create a New Virtual Machine.
38. Enter a clear VM name such as csa15-localvm-yourregno.
39. Select Linux as the type and the closest version available.
40. Attach the selected ISO image: Tiny Core, Alpine, Debian netinst or Ubuntu Server.
41. Allocate CPU based on CO2 AT1 sizing, but keep within safe host limits.
42. Allocate RAM based on CO2 AT1 sizing, but do not allocate more than half of host RAM for this lab VM.
43. Create a virtual disk. For Tiny Core or Alpine, 2-8 GB is enough for evidence. For Debian/Ubuntu, use 15-25 GB when storage permits.
44. Finish creation and start the VM.

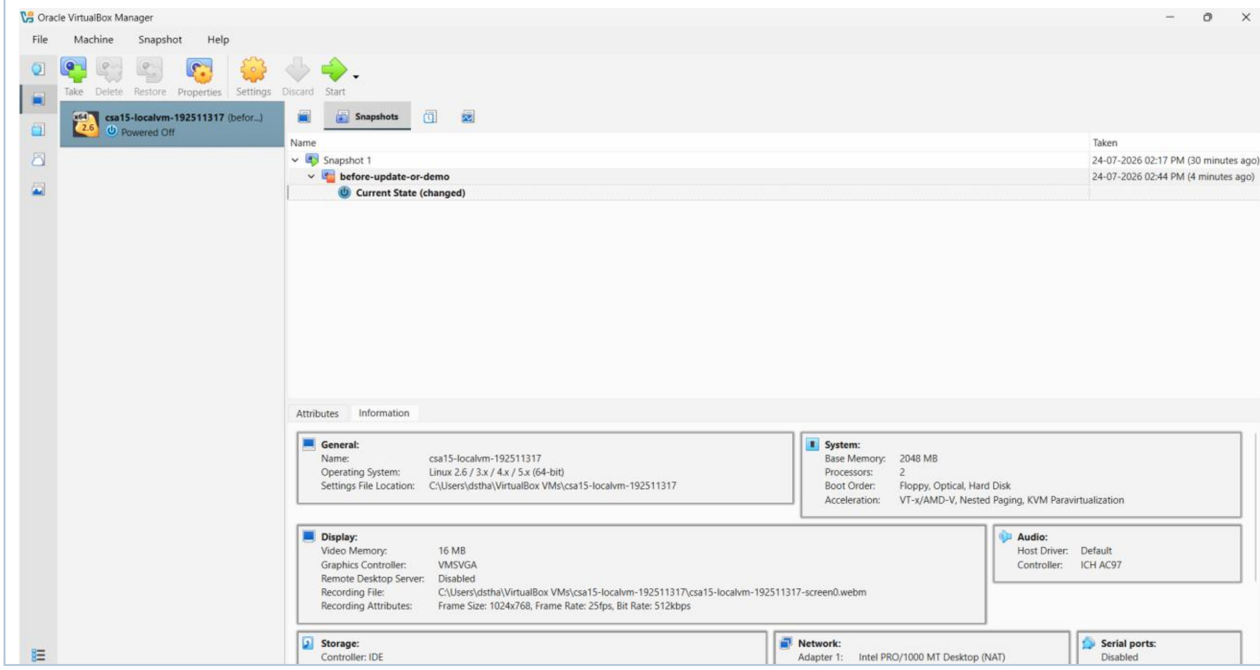
VM Setting	Recommended Lab Range	Student Value
VM name	csa15-localvm-regno	csa15-localvm-192511317
OS image	Tiny Core / Alpine / Debian / Ubuntu	Alpine Linux virt 3.24.1

vCPU	1-2 for lightweight OS; 2+ if host allows	2
RAM	512 MB-2 GB lightweight; 2-4 GB Ubuntu	2048MB
Virtual disk	2-8 GB lightweight; 15-25 GB Debian/Ubuntu	8GB
Network mode	NAT for normal lab use	NAT
Display / boot mode	Default unless OS requires change	Defaut

Host Safety Guidance

Do not allocate all host resources to the VM. Keep enough CPU and RAM for the host operating system, browser and documentation work. If the host has 8 GB RAM, a 1 GB or 2 GB VM is more appropriate than a 6 GB VM.

Evidence Space: Paste VM settings screenshot before first boot



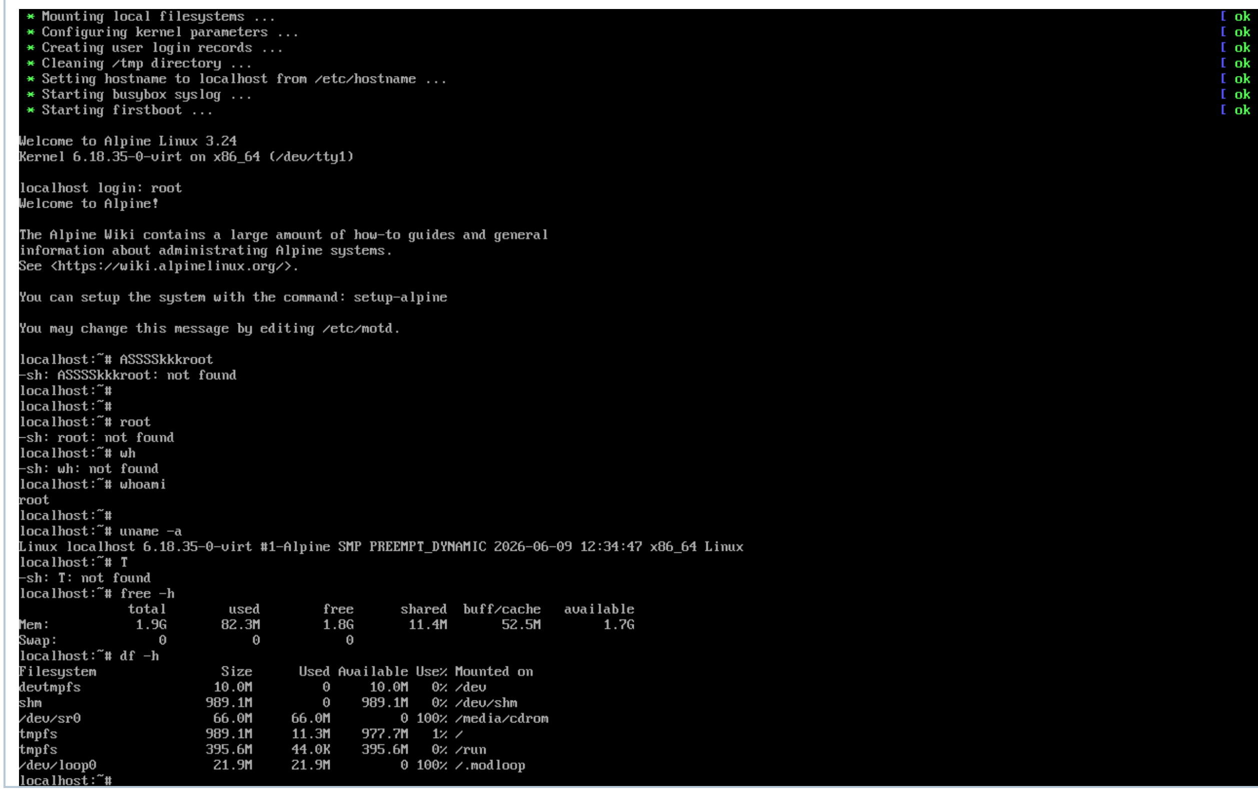
C. Boot and Verify the VM

45. Start the VM and wait for the OS boot screen.
46. For Tiny Core, confirm that the graphical desktop appears.
47. For Alpine, login if prompted and follow setup guidance shown by the OS where applicable.
48. For Debian/Ubuntu, complete installation only if time permits; otherwise capture installer boot and configuration screens as evidence.
49. Open terminal inside the VM where possible.
50. Run or record equivalent commands: uname -a, free -h, df -h and ip addr.

Checkpoint	Status	Evidence / Observation
VM starts without boot error.	[X] Yes [ ] Rework	Alpine Linux 3.24 boot sequence completed cleanly, reaching the login prompt with no errors
OS screen or terminal is visible.	[x] Yes [ ] Rework	Boot text and localhost login: prompt clearly visible in VM window

CPU/RAM/disk allocation is verified.	[x] Yes [ ] Rework	free -h showed 1.9G total RAM (matches 2GB allocation); df -h showed disk mounts consistent with the 8GB virtual disk
Network mode is recorded.	[x] Yes [ ] Rework	NAT confirmed; ip addr showed eth0 with inet 10.0.2.15/24 obtained via udhcpc from VirtualBox's NAT gateway

Evidence Space: Paste VM boot/login/terminal screenshots



```
 * Mounting local filesystems ...                                   [ ok ]
 * Configuring kernel parameters ...                                [ ok ]
 * Creating user login records ...                                  [ ok ]
 * Cleaning /tmp directory ...                                      [ ok ]
 * Setting hostname to localhost from /etc/hostname ...             [ ok ]
 * Starting busybox syslog ...                                      [ ok ]
 * Starting firstboot ...                                           [ ok ]

Welcome to Alpine Linux 3.24
Kernel 6.18.35-0-virt on x86_64 (/dev/tty1)

localhost login: root
Welcome to Alpine!

The Alpine Wiki contains a large amount of how-to guides and general
information about administrating Alpine systems.
See <https://wiki.alpinelinux.org/>.

You can setup the system with the command: setup-alpine

You may change this message by editing /etc/motd.

localhost:~# ASSSSkkkroot
-sh: ASSSSkkkroot: not found
localhost:~#
localhost:~#
localhost:~# root
-sh: root: not found
localhost:~# wh
-sh: wh: not found
localhost:~# whoami
root
localhost:~#
localhost:~# uname -a
Linux localhost 6.18.35-0-virt #1-Alpine SMP PREEMPT_DYNAMIC 2026-06-09 12:34:47 x86_64 Linux
localhost:~# T
-sh: T: not found
localhost:~# free -h
              total        used        free      shared  buff/cache   available
Mem:           1.9G       82.3M        1.8G       11.4M       52.5M        1.7G
Swap:             0           0           0
localhost:~# df -h
Filesystem                Size      Used Available Use% Mounted on
devtmpfs                 10.0M         0     10.0M   0% /dev
shm                     989.1M         0    989.1M   0% /dev/shm
/dev/sr0                 66.0M     66.0M         0 100% /media/cdrom
tmpfs                   989.1M     11.3M    977.7M   1% /
tmpfs                   395.6M     44.0K    395.6M   0% /run
/dev/loop0               21.9M     21.9M         0 100% /.modloop
localhost:~#
```

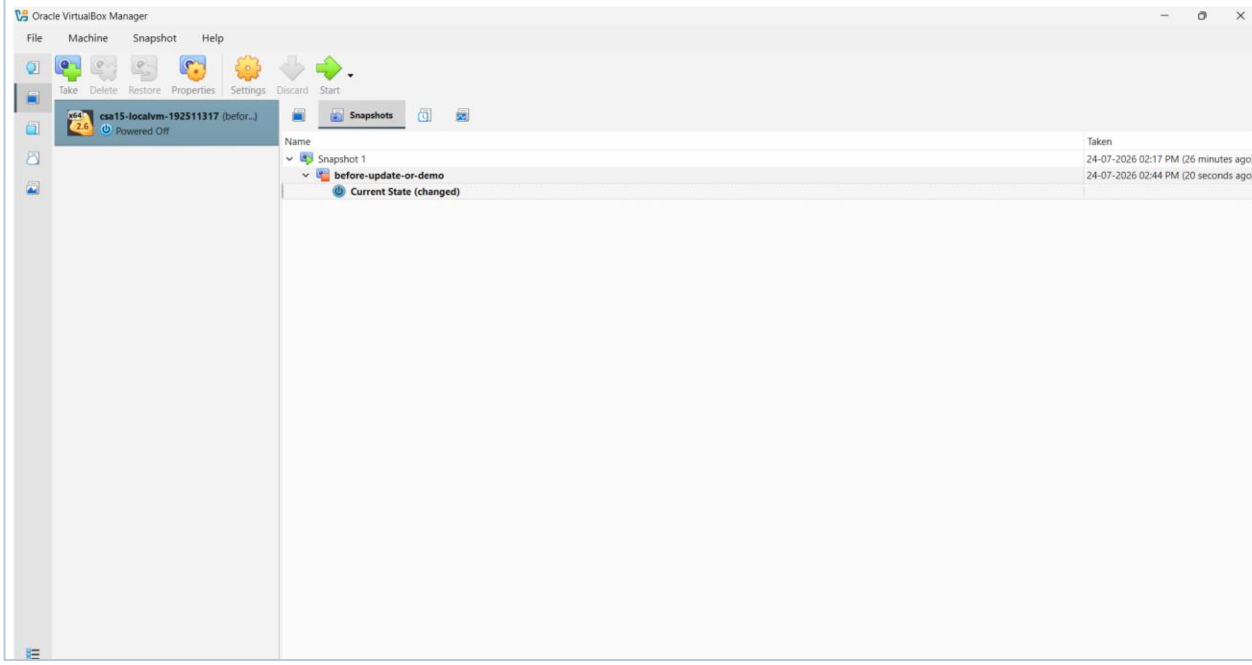
D. Snapshot and Clone Practice

51. Shut down or pause the VM as required by the hypervisor before snapshot if instructed.
52. Create a snapshot named before-update-or-demo.
53. Record snapshot name, date and reason.
54. Create a linked clone or full clone if the device has enough storage. If not, write the reason and create a clone plan.
55. Start the VM again and confirm it still boots.

Recovery Item	Student Evidence / Decision
Snapshot name	before-update-or-demo
Snapshot timing	Taken after clean boot and network verification, before any package installation or system changes
Snapshot reason	Provides a rollback point in case a future update, package install, or configuration change breaks the VM
Clone attempted?	Yes **/ No**
Clone type	Full / Linked **/ Not attempted**

Storage impact observed	Snapshot uses a differencing disk, adding minimal storage overhead since no changes have been made since it was taken

Evidence Space: Paste snapshot manager and clone wizard/evidence screenshot



E. Local VM Security Reflection

Security Area	Student Decision
VM network mode	<u>**NAT**</u> / Bridged / Host-only / Other
Why this mode is selected	NAT gives the VM outbound internet access for testing (e.g. running udhcpc) without exposing it to inbound connections from the host's network — safest default for a temporary lab VM
Guest password policy	Root login with no password (Alpine's live-boot default) — acceptable here only because the VM stays on NAT and is never exposed externally
Shared folders enabled?	Yes **/** <u>**No**</u>
Clipboard/drag-drop enabled?	Yes / <u>**No**</u>
Risk if VM is exposed to public network	With no root password set, a public/bridged exposure would allow anyone reaching the VM's IP to log in as root with no authentication — a critical risk. NAT mode prevents this by keeping the VM unreachable from outside the host.

Part 5: Track C - Docker Container Practical and Comparison

Docker Desktop: https://www.docker.com/products/docker-desktop/

This path helps students understand how a container differs from a VM. A VM virtualizes hardware and runs a guest OS. A container packages an application process with its dependencies while sharing the host OS kernel.

Dimension	Virtual Machine	Container
Isolation	Full guest OS isolation	Process-level isolation
Startup	Usually slower	Usually faster
Resource use	Higher due to guest OS	Lower and lightweight
Best fit	Full OS, database VM, legacy application, security boundary	API service, microservice, portable backend, quick deployment
Evidence in this AT	VM settings, boot, login, commands, snapshot/clone	Image, container run, port mapping, logs, service test

A. Container Verification Steps

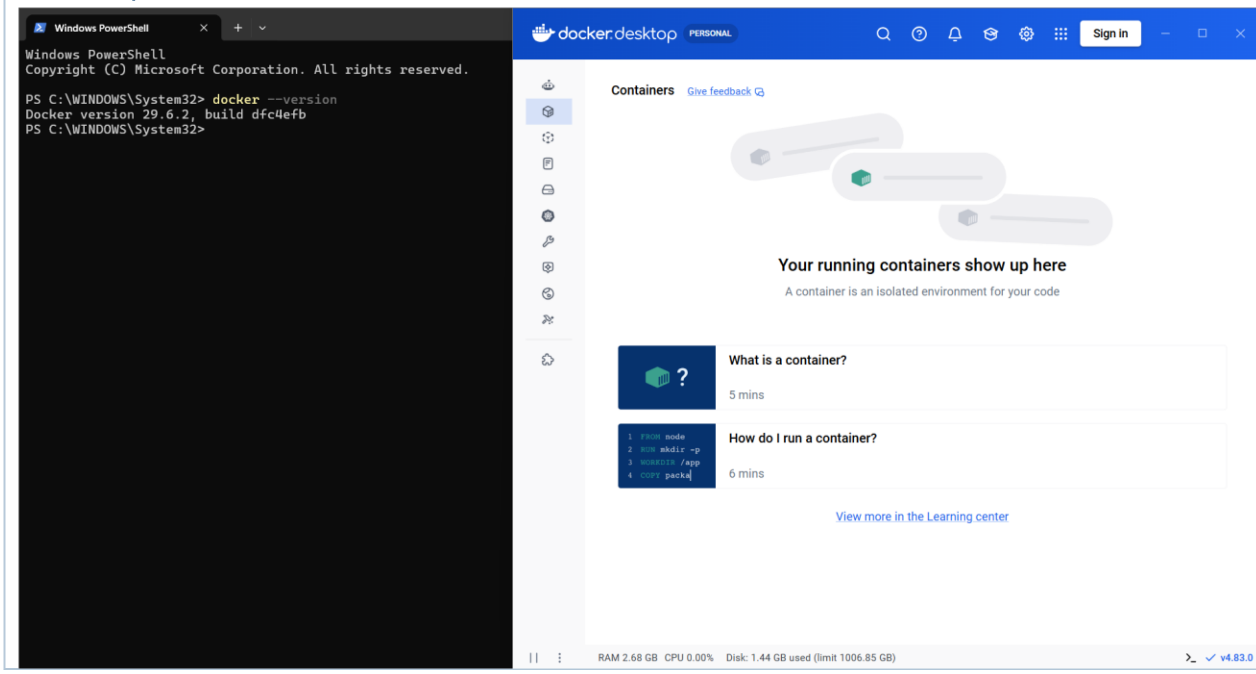
56. Install Docker Desktop from the official site if allowed on the device.
57. Open Terminal or PowerShell and run: docker --version.
58. Run a lightweight container such as: docker run --rm hello-world.
59. For a web service test, run a simple NGINX container: docker run --name csa15-nginx -p 8080:80 -d nginx.
60. Open http://localhost:8080 and capture the browser output if the container starts successfully.
61. Check logs using: docker logs csa15-nginx.
62. Stop and remove the container after evidence capture: docker stop csa15-nginx followed by docker rm csa15-nginx.

Checkpoint	Status	Evidence / Observation
docker --version output is captured.	[x] Yes [ ] Rework	Docker version 29.6.2, build dfc4efb
A container run command is captured.	[x] Yes [ ] Rework	docker run --name csa15-nginx -p 8080:80 -d nginx ran successfully, returning a container ID
Port mapping or service evidence is captured if web test is used.	[x] Yes [ ] Rework	docker ps showed 0.0.0.0:8080->80/tcp; browser confirmed NGINX welcome page at localhost:8080
Container logs or status are captured.	[x] Yes [ ] Rework	docker logs csa15-nginx showed startup messages plus real access log entries (GET / 200)
Container is stopped/removed after evidence capture.	[x] Yes [ ] Rework	docker stop and docker rm both confirmed by name; docker ps -a showed an empty list afterward

Docker Evidence	Student Observation
Docker version	29.6.2, build dfc4efb
Image used	nginx (latest)
Container name	csa15-nginx
Port mapping	8080 (host) → 80 (container)
Service URL tested	http://localhost:8080

Log/status observation	Startup logs showed successful config; access logs recorded GET / 200 (page load success) and GET /favicon.ico 404 (normal automatic browser request, not an error)
Stop/remove evidence	docker stop csa15-nginx and docker rm csa15-nginx both returned the container name confirming success; docker ps -a confirmed no remaining containers

Evidence Space: Paste Docker command and browser/service screenshots



B. Capstone Module Decision

Capstone Module	VM / Container / Managed Service	Technical Reason
Frontend	Managed Service	Static site served via CDN hosting (e.g. Firebase Hosting) — no server management needed
Backend/API	Serverless Function	Requests are short and spiky, matching pay-per-invocation compute rather than an always-on server
Database	Managed Service	Fully managed NoSQL store (Firestore) with built-in scaling and replication
File storage	Managed Service	Purpose-built object storage for screenshots — cheaper and more durable than VM/container disk
AI/Python module	Container	OCR/ELA image processing can exceed serverless function memory/time limits under heavier load, so a container gives predictable resource allocation

Analytics/dashboard	Serverless Function	Aggregation runs periodically via a scheduled function, not continuously
Notification/background job	Serverless Function	Event-driven alerts fire only when triggered, matching serverless billing

C. Optional Container and Orchestration Extension

Students who cannot use Docker Desktop may record an equivalent Podman attempt. Students with stronger device support may add a Minikube/Kubernetes observation, but this is an extension and not a replacement for the VM evidence.

Podman: https://podman.io/

Minikube: https://minikube.sigs.k8s.io/docs/start/

Kubernetes documentation: https://kubernetes.io/docs/home/

Optional Tool	Possible Evidence	Student Observation
Podman	podman --version, image pull/run, logs, stop/remove evidence	Not attempted — Docker Desktop was successfully used instead, covering the required container evidence
Minikube	minikube start/status, kubectl get nodes/pods evidence	Not attempted
Kubernetes	Simple deployment/service YAML or command evidence if attempted	Not attempted

Part 6: Track D - Technical Simulation Path

Use Only When Required

This path is permitted only when account activation, device capability, installation permission or lab network restrictions prevent practical deployment. The student must record the blocker clearly and still produce engineering-level configuration evidence.

CloudSim: https://github.com/Cloudslab/cloudsim

Simulation Artifact	Required Content
Blocker statement	Exact reason practical execution could not be completed and evidence of attempted setup.
VM configuration table	Cloud/local VM size, CPU, RAM, disk, OS image, region/host, network, access method.
Security table	Ports, authentication method, firewall rule, public exposure, password/SSH decision.
Snapshot/clone plan	Names, timing, retention and rollback purpose.
Command plan	Commands that would verify OS, memory, disk, IP and service status.
Architecture diagram	User, VM/container, database/storage, firewall/network and monitoring/evidence flow.
Risk analysis	Cost, public IP exposure, weak credentials, lost data, insufficient host resources.

Blocker Evidence and Justification

no blocker was encountered

Planned Configuration	Student Value
Platform	Azure / VirtualBox / VMware / Docker
OS image	
vCPU	
RAM	
Disk	
Network mode / firewall	
Access method	
Service to verify	

Paste or draw the simulation architecture diagram here

Part 7: CO2 Reflection and Infrastructure Defense

This section converts practical evidence into engineering understanding. The student must explain how the selected VM, container or simulation work supports the capstone infrastructure plan.

Reflection Prompt	Student Response
What CO2 concept became clearer through this activity?	The practical difference between a VM and a container became concrete rather than theoretical — the Alpine VM took noticeably longer to boot and reserved a fixed 2GB of RAM whether used or not, while the Docker NGINX container started in under a second and only used resources while actively running.
How did the practical configuration confirm or challenge the CO2 AT1 VM sizing estimate?	CO2 AT1 estimated 2-4 vCPU and 12GB RAM for the workload. The Azure VM (B2s_v2: 2 vCPU, 8GB RAM) landed close to that estimate and proved sufficient for running and testing a web service, confirming the sizing logic was reasonable even though the exact available Azure SKU differed slightly from the original numbers.
Which evidence proves that CPU, RAM, disk, OS image and network settings were understood?	Verification commands (uname -a, free -h, df -h, ip addr) were run on both the local VM and the Azure VM, and their output was checked against the values actually configured (e.g. 8GiB RAM shown by free -h matching the B2s_v2 selection), confirming each setting was intentional rather than assumed.
Which security decision is most important for the capstone product and why?	Restricting inbound access to SSH only by default, with port 80 opened temporarily and closed immediately after testing, matters most for PulsePay specifically — since the product handles screenshots containing partial financial/UPI details, minimizing the attack surface on any compute resource is the highest priority.
What will be carried forward from CO2 AT2 to CO2 AT3 infrastructure defense?	The Azure VM configuration and NSG port rules, the snapshot naming and timing convention, and the VM-vs-container decisions made per capstone module (Part 5-B) will all carry forward as the baseline for the CO2 AT3 infrastructure design defense.

Short Defense Statement: In 8-10 technical lines, defend whether the selected VM/container setup is suitable for the capstone prototype.

This practical demonstrated three virtualization approaches for the same service: a local Alpine VM, a Docker NGINX container, and a real Azure Ubuntu VM (B2s_v2, 2 vCPU/8GB). All three were configured, started, and verified using the same core commands, confirming consistent understanding of OS, memory, disk, and network layers across environments. The Azure VM used SSH-key-only authentication and kept inbound access to port 22 by default, opening port 80 only temporarily for the NGINX test before closing it again — appropriate given PulsePay's sensitivity around screenshot data. Auto-shutdown was configured as an additional cost-control safeguard. The Docker comparison showed containers start faster and use fewer resources than a full VM, directly informing the decision to run PulsePay's AI/ELA module in a container rather than a VM. This setup is suitable for prototype-stage evidence; a production deployment would move most components to fully serverless/managed services, as already planned in CO2 AT1.

Student-Facing Assessment Criteria

Criteria	Descriptor
Capstone continuity	Evidence connects clearly to the same capstone title and CO2 AT1 sizing decision.
Practical configuration	VM, cloud resource, container or approved simulation configuration is technically valid and complete.
Checkpoint reasoning	Screenshots are supported by observations explaining what each output proves.
Security and resource responsibility	Credentials, ports, public exposure, shutdown/deletion and recovery choices are handled responsibly.
Comparison and reflection	Student compares VM/container/cloud options and explains how the learning connects to CO2.
Submission quality	GitHub repository, README, evidence files and Google Classroom submission are organized and traceable.

Part 8: Evidence Index, GitHub and Google Classroom Submission

The submission must be traceable. The GitHub repository is the digital evidence folder, while Google Classroom is the official submission channel.

Repository Item	Expected Content
README.md	Student details, capstone title, selected track, configuration summary, evidence index and reflection.
Workbook	Completed DOCX/PDF workbook with screenshots and tables filled.
Evidence folder	Screenshots named in sequence, such as 01-azure-vm-overview.png.
Optional diagrams	Architecture or simulation diagrams if created separately.
Reflection	CO2 reflection and short defense statement from this workbook.

Evidence No.	File / Screenshot Name	What It Proves
1	01-vm-created.png	Local VM configured (name, RAM, CPU, disk) before boot
2	02-vm-boot-verification.png	Local VM OS/kernel, RAM, disk verified via terminal
3	03-network-verification.png	Local VM network (NAT) working, IP obtained via udhcpc
4	04-snapshot-created.png	Local VM snapshot "before-update-or-demo" created
5	05-docker-version.png	Docker Desktop installed and functional
6	06-docker-nginx-running.png	Container running with correct port mapping (8080→80)
7	07-nginx-browser-test.png	Docker-hosted NGINX reachable in browser

8	08-docker-logs.png	Container serving real HTTP requests, logs inspected
9	09-docker-cleanup.png	Container stopped and removed after evidence capture
10	10-azure-vm-overview.png	Azure VM deployed successfully, Running status, public IP visible
11	11-azure-ssh-connected.png	SSH connection to Azure VM established via key auth
12	12-azure-verification-commands.png	Azure VM OS/RAM/disk/network verified via terminal
13	13-azure-nginx-installed.png	NGINX installed and active on Azure VM
14	14-azure-nginx-browser-test.png	Azure-hosted NGINX reachable publicly in browser
15	15-azure-port80-closed.png	Temporary HTTP port closed, SSH-only access restored
16	16-azure-vm-final-status.png	Final VM status confirmed Running, with Auto-shutdown enabled

Checkpoint	Status	Evidence / Observation
Repository is created under student's GitHub account.	[x] Yes [ ] Rework	Repository created and evidence uploaded
README.md is complete.	[x] Yes [ ] Rework	README includes student details, capstone title, tracks completed, and evidence index summary
Workbook is uploaded.	[x] Yes [ ] Rework	Completed DOCX/PDF workbook uploaded to repository
Screenshots are uploaded and named clearly.	[x] Yes [ ] Rework	All 16 evidence files uploaded in /evidence, named in sequence
Google Classroom submission contains the GitHub repository link.	[x] Yes [ ] Rework	Link submitted via Google Classroom
No passwords, private keys, tokens or sensitive account details are uploaded.	[x] Yes [ ] Rework	.pem private key file kept locally only, not uploaded; no credentials appear in any screenshot

Student Assurance

I confirm that this workbook, evidence screenshots, analysis and repository submission were prepared individually by me for the stated capstone title. I have not uploaded passwords, tokens, private keys or sensitive account information.

Student Name	D.S.THARUNIYA PRIYA

Register Number	192511317
Signature	D.S.Tharuniya Priya
Date	29/07/2026

Faculty Verification

Faculty Name	Dr C. Rajesh Babu
Employee ID	SSETSCS415
Constructive Feedback / Remarks	
Strength Observed	
Improvement Suggested	
Signature / Date	

Reference Links Used in This Workbook

Azure for Students: https://azure.microsoft.com/en-us/free/students

Azure for Students offer details: https://azure.microsoft.com/en-us/pricing/offers/ms-azr-0170p

Azure Linux VM quickstart: https://learn.microsoft.com/en-us/azure/virtual-machines/linux/quick-create-portal

VirtualBox downloads: https://www.virtualbox.org/wiki/Downloads

VMware Workstation and Fusion: https://www.vmware.com/products/desktop-hypervisor/workstation-and-fusion

Tiny Core Linux downloads: https://distro.ibiblio.org/tinycorelinux/downloads.html

Alpine Linux downloads: https://alpinelinux.org/downloads/

Alpine virt ISO directory: https://dl-cdn.alpinelinux.org/alpine/latest-stable/releases/x86_64/

Debian netinst: https://www.debian.org/download

Debian network install: https://www.debian.org/CD/netinst/

Ubuntu Server: https://ubuntu.com/download/server

Docker Desktop: https://www.docker.com/products/docker-desktop/

Podman: https://podman.io/

Minikube: https://minikube.sigs.k8s.io/docs/start/

Kubernetes: https://kubernetes.io/docs/home/

CloudSim: https://github.com/Cloudslab/cloudsim